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A semilinear heat equation with variable reaction and singular initial data

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ABSTRACT

We study the semilinear parabolic problem $u_t - \Delta u = f(u)^{p(x)}$ in $\Omega \times (0, T)$ with zero Dirichlet condition on the boundary of a smooth bounded domain $\Omega \subset \mathbb{R}^N$. We assume that $p \in C(\Omega, \mathbb{R}_+)$ is continuous satisfying $0 < p^- = \min_{x \in \Omega} p(x) \leq p(x) \leq \max_{x \in \Omega} p(x) = p^+ < \infty$, $f \in C([0, +\infty))$ is non-decreasing, and the initial data are in $L^r(\Omega)$, $1 \leq r < \infty$. We prove the existence, non-existence and uniqueness of solution in the space $L^\infty((0, T), L^r(\Omega))$. In our approach we use the supersolution method which allows us to conclude that this solution belongs to the space $C([0, T], L^r(\Omega)) \cap L^\infty_{loc}((0, T), L^\infty(\Omega))$.

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1. Introduction

Let $\Omega \subset \mathbb{R}^N$ be a bounded domain with smooth boundary $\partial\Omega$, $T > 0$ and $\mathbb{R}_+ = (0, +\infty)$. We study the following parabolic problem with a variable non-linearity

$$\begin{cases} u_t - \Delta u = f(u)^{p(x)} & \text{in } \Omega \times (0, T), \\ u = 0 & \text{on } \partial\Omega \times (0, T), \\ u(0) = u_0 \geq 0 & \text{in } \Omega, \end{cases} \quad (1)$$

where $f \in C([0, +\infty))$ is non-decreasing, $p \in C(\Omega, \mathbb{R}_+)$ verifies

$$0 < p^- \leq p(x) \leq p^+ < \infty, \quad \text{for all } x \in \Omega, \quad (2)$$

with $p^- = \min_{x \in \Omega} p(x)$ and $p^+ = \max_{x \in \Omega} p(x)$. The initial data are in $L^r(\Omega)$, $1 \leq r < \infty$, and we are concerned only with non-negative solutions.

Problem (1) appears in several models of applied sciences such as electrorheological fluids (see [1–7]), thermorheological fluids (see [8]), image processing (see [9–11]), magnetostatics (see [12]), chemical reactions, heat transfer, and population dynamics (see [13–15]).

Problem (1), with continuous initial data, has been considered in [16] (for $f \in C([0, +\infty))$), [17] (for $f(t) = e^t$), [18] (for $f(t) = e^t$ and $p(x) = 1$) and [19–21] (for $f(t) = t$), and with initial data in

$L^r(\Omega)$ in [22–28] (for $f(t) = t$ and $p(x) = p > 1$ constant). Other problems considering p depending on x have recently been treated in [29–31].

When $p(x) = 1$ is constant, we have the following problem

$$\begin{cases} u_t - \Delta u = f(u) & \text{in } \Omega \times (0, T), \\ u = 0 & \text{on } \partial\Omega \times (0, T), \\ u(0) = u_0 \geq 0 & \text{in } \Omega, \end{cases} \quad (3)$$

which has been studied in [32, Theorem 3.4 and Corollary 4.5]. More explicitly, the following was shown.

Theorem 1.1 ([32]): *Let $f \in C([0, +\infty))$ be non-decreasing and $1 \leq r < \infty$. The problem (3) has a L^r -solution with initial data $u_0 \in L^r(\Omega)$, $u_0 \geq 0$ if and only if*

$$\begin{aligned} \limsup_{t \rightarrow +\infty} t^{-p^*} f(t) &< \infty \quad \text{for } r > 1, \\ \int_1^{+\infty} \sigma^{-p^*} F(\sigma) d\sigma &< \infty \quad \text{for } r = 1, \end{aligned}$$

where $F(\sigma) = \sup_{1 \leq t \leq \sigma} f(t)/t$ and

$$p^* = 1 + \frac{2r}{N}. \quad (4)$$

Remark 1.2: It is worth mentioning that the L^r -solution found in [32] is in the class $L^\infty((0, T), L^r(\Omega))$. We show here, see Proposition 2.8, that this solution belongs to the class

$$\left\{ u \in C([0, T], L^r(\Omega)) \cap L_{loc}^\infty((0, T), L^\infty(\Omega)); \sup_{t \in (0, T]} t^{N/2r} \|u(t)\|_{L^\infty} < \infty \right\},$$

which coincides with the results obtained in [22] and [28] for $f(t) = t^p$ with $p > 1$.

The main purpose of this work is to extend the results obtained in [32] for problem (1) with p a function depending on x . As far as we know, there are no works that deal with the existence of L^r -solutions to the problem (1) with initial data in Lebesgue spaces. As a consequence of our results, we are able to completely analyze the existence of L^r -solutions (in the sense of Definition 2.2) of equation $u_t - \Delta u = u^{p(x)}$ in $\Omega \times (0, T)$ (see Corollaries 1.5 and 1.8).

In our first result, we analyze the existence of non-negative L^r -solutions.

Theorem 1.3: *Let $f \in C([0, +\infty))$ be non-decreasing, $p \in C(\Omega, \mathbb{R}_+)$ verifying (2), p^* given by (4), $F_1(\sigma) = \sup_{1 \leq t \leq \sigma} f(t)^{p^+}/t$, and $F_2(\sigma) = \sup_{1 \leq t \leq \sigma} f(t)^{p^-}/t$.*

(i) *If*

$$\limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^+} < \infty \quad \text{and} \quad \limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^-} < \infty, \quad \text{for } r > 1, \quad (5)$$

or

$$\int_1^{+\infty} \sigma^{-p^*} F_1(\sigma) d\sigma < \infty \quad \text{and} \quad \int_1^{+\infty} \sigma^{-p^*} F_2(\sigma) d\sigma < \infty, \quad \text{for } r = 1, \quad (6)$$

then problem (1) has a L^r -solution defined on an interval $[0, T]$ for every $u_0 \in L^r(\Omega)$, with $u_0 \geq 0$. Moreover, $u \in C([0, T], L^r(\Omega))$ and there exists a constant $C > 0$ such that $\|u(t)\|_{L^\infty} \leq Ct^{-N/2r}$ for $t \in (0, T]$.

(ii) If

$$\limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^-} = +\infty, \quad \text{for } r > 1,$$

or

$$\int_1^{+\infty} \sigma^{-p^*} F_2(\sigma) d\sigma = +\infty, \quad \text{for } r = 1,$$

then there exist $u_0 \in L^r(\Omega)$, $u_0 \geq 0$ such that problem (1) does not have a L^r -solution.

Remark 1.4: Theorem 1.3 is obtained using the same ideas of [32] with small adaptations. Thus, this result also holds when $\Omega = \mathbb{R}^N$ considering the additional assumptions $\limsup_{t \rightarrow 0} f(t)^{p^+}/t < \infty$ and $\limsup_{t \rightarrow 0} f(t)^{p^-}/t < \infty$ in (5) and (6).

As a consequence of Theorem 1.3 we have.

Corollary 1.5: Suppose that $f(t) = t$ for $t \geq 0$, and $p \in C(\Omega, \mathbb{R}_+)$ verifying (2).

- (i) If $p^+ < p^*$ (resp. $p^+ = p^*$) then problem (1) has a L^r -solution for every $u_0 \in L^r(\Omega)$, $u_0 \geq 0$ with $1 \leq r < \infty$ (resp. $r > 1$).
- (ii) If $p^- > p^*$ (resp. $p^- = p^*$) then there exists $u_0 \in L^r(\Omega)$, $u_0 \geq 0$ such that problem (1) does not admit a L^r -solution with $1 \leq r < \infty$ (resp. $r = 1$).

Note that there are gaps in Theorem 1.3. For example, for $f(t) = t$ the theorem does not work when: $p^- < p^* < p^+$, $r \geq 1$; $p^* = p^+$, $r = 1$; and $p^* = p^-$, $r > 1$ (see Corollary 1.5). Thus, in the next results (Theorem 1.6 and Corollary 1.8) we focus in these cases and we show that the existence and non-existence of L^r -solutions might occur simultaneously. To do this, we consider initial data in the sets $\mathcal{I}^\rho(x_0)$ and $\mathcal{I}_\rho(x_0)$ defined by

$$\begin{aligned} \mathcal{I}^\rho(x_0) = \{ \psi \in L^r(\Omega); \psi \geq 0 \quad \text{and} \quad |x - x_0|^\rho \psi^r(x) \leq K \chi_{B_a(x_0)}(x) \quad \text{a.e. in } \Omega, \\ \text{for some } K, a > 0 \}, \end{aligned}$$

$$\begin{aligned} \mathcal{I}_\rho(x_0) = \{ \psi \in L^r(\Omega); \psi \geq 0 \quad \text{and} \quad |x - x_0|^\rho \psi^r(x) \geq K \chi_{B_a(x_0)}(x) \quad \text{a.e. in } \Omega, \\ \text{for some } K, a > 0 \}, \end{aligned}$$

where $B_a(x_0) \subset \Omega$ is the open ball with radius $a > 0$ centered at x_0 , for $0 < \rho < N$ and $r \geq 1$. The condition $0 < \rho < N$ assures that $|\cdot - x_0|^{-\rho/r} \chi_{B_a(x_0)} \in L^r(\Omega)$. Here $\chi_{B_a(x_0)}$ denotes the characteristic function of the ball $B_a(x_0)$. When $x_0 = 0 \in \Omega$, we denote $\mathcal{I}^\rho(0) = \mathcal{I}^\rho$ and $\mathcal{I}_\rho(0) = \mathcal{I}_\rho$ and $B_a(0) = B_a$.

The sets $\mathcal{I}^\rho$ and $\mathcal{I}_\rho$ were used in [33] to show the existence of a second critical value $\rho^* = 2r/(p - 1)$, when $f(t) = t$ and $p(x) = p > 1$ is a constant function.

Considering $u_0 \in \mathcal{I}^\rho(x_0)$ and $u_0 \in \mathcal{I}_\rho(x_0)$ we obtain the following new result.

Theorem 1.6: Let $f \in C([0, +\infty))$ be non-decreasing, convex and $f(0) = 0$, $p \in C(\Omega, \mathbb{R}_+)$ verifying (2) and p^* given by (4).

- (i) Assume that there exists $p_0 \geq p^*$ such that

$$\limsup_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^+} < \infty, \quad \limsup_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^-} < \infty, \quad \text{and } p^- \geq 1.$$

Then, for every $x_0 \in \Omega$ and $u_0 \in \mathcal{I}^\rho(x_0)$, with $0 < \rho < 2r/(p_0 - 1) \leq N$, the problem (1) has a non-negative L^r -solution on some interval $(0, T)$. Moreover, $u \in C([0, T], L^r(\Omega)) \cap L_{loc}^\infty((0, T), L^\infty(\Omega))$ and $\|u(t)\|_{L^\infty} \leq Ct^{-\rho/2r}$ for all $t \in (0, T)$ and some constant $C > 0$.

(ii) Assume that there exists $p_0 > p^*$ such that

$$\liminf_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^+} > 0, \quad \text{with } p^+ > 1,$$

or

$$\liminf_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^-} > 0 \quad \text{with } p^- \geq 1.$$

Then there exists $x_0 \in \Omega$ such that for every $u_0 \in \mathcal{I}_\rho(x_0)$, with $2r/(p_0 - 1) < \rho < N$, the problem (1) does not have a non-negative L^r -solution.

Remark 1.7: Here are some comments about Theorem 1.6.

- (a) Some convex and non-decreasing functions where it is possible to apply Theorem 1.6 are: $f(t) = t^q$ and $f(t) = (1 + t)(\ln(1 + t))^q$ for $q > 1$.
- (b) In item (i), the convexity of f , $f(0) = 0$, and $p^- \geq 1$ are necessary to ensure that the function $t \in (0, +\infty) \mapsto [f(t)^{p^+} + f(t)^{p^-}]/t$ is well defined and non-decreasing. Without these assumptions, it is possible to obtain the same conclusion assuming $u_0 \in \mathcal{I}^\rho(x_0)$ with $\rho < N$ and

$$\int_1^{+\infty} \sigma^{-(1+2r/\rho)} F_i(\sigma) d\sigma < \infty, \quad i = 1, 2, \quad (7)$$

where $F_1(\sigma) = \sup_{1 \leq \tau \leq \sigma} f(\tau)^{p^+}/\tau$ and $F_2(\sigma) = \sup_{1 \leq \tau \leq \sigma} f(\tau)^{p^-}/\tau$ for $\sigma \geq 1$, see Remark 3.1.

- (c) The convexity of f is needed in item (ii) because we use Proposition 2.6.
- (d) The assumptions of item (ii) imply that $\limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^+} = +\infty$ which is not covered by Theorem 1.3. We prove the claim only when $\liminf_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^+} > 0$ because the other case is similar. In fact, we argue by contradiction and assume that $\limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^+} < \infty$. Therefore, $f(t)^{p^+} \leq Ct^{p^*}$ for t sufficiently large and some constant $C > 0$. Thus, $t^{-p_0} f(t)^{p^+} \leq Ct^{p^* - p_0}$ for t large and we have a contradiction letting $t \rightarrow +\infty$.

When p is a constant function, that is, $p(x) = p^- = p^+$ ($x \in \Omega$), it is possible to apply Theorem 1.3. Thus, assuming $p^- < p^+$ we obtain.

Corollary 1.8: Suppose that $f(t) = t$, $t \geq 0$, $p \in C(\Omega, \mathbb{R}_+)$ verifying (2), and $p^- < p^+$.

- (i) If $1 \leq p^- < p^* \leq p^+$, then problem (1) has a L^r -solution for every $x_0 \in \Omega$ and $u_0 \in \mathcal{I}^\rho(x_0)$, with $0 < \rho < 2r/(p_0 - 1)$ for every $p_0 \geq p^+$ and $p_0 > p^*$.
- (ii) If $p^- \leq p^* < p^+$, then there exists $x_0 \in \Omega$ such that for every $u_0 \in \mathcal{I}_\rho(x_0)$, with $2r/(p_0 - 1) < \rho < N$ and $p^+ \geq p_0 > p^*$, the problem (1) does not admit a L^r -solution.

Remark 1.9: As it was mentioned before, the Corollary 1.5 does not cover the cases: $p^- < p^* < p^+$, $r \geq 1$; $p^* = p^+$, $r = 1$; and $p^* = p^-$, $r > 1$, when $f(t) = t$, $t \geq 0$. These cases are analyzed in Corollary 1.8.

Results about the uniqueness for $f(t) = t$ with $p(x) = p > 1$ a constant function have been obtained in [22, 26, 28, 33, 34]. As in [33, 34] we assume that $f \in C([0, +\infty))$ is locally Lipschitz

and define $\mathcal{G} : [0, +\infty) \rightarrow [0, +\infty)$ given by

$$\mathcal{G}(t) = \sup_{\substack{0 \leq u, v \leq t \\ u \neq v}} \frac{f(u) - f(v)}{u - v}, \quad \text{for } t > 0, \mathcal{G}(0) = 0.$$

Our uniqueness result is the following.

Theorem 1.10: *Let $f \in C([0, +\infty))$ be locally Lipschitz, non-decreasing, and $p \in C(\Omega, \mathbb{R}_+)$ verifying (2) with $p^- \geq 1$. Problem (1) admits a unique L^r -solution in the class*

$$\left\{ u \in C([0, T], L^r(\Omega)) \cap L_{loc}^\infty((0, T), L^\infty(\Omega)); \sup_{t \in (0, T)} t^{\alpha/2r} \|u(t)\|_{L^\infty} < \infty \right\}, \quad (8)$$

for $\alpha > 0$ if the following integral condition

$$\int_1^{+\infty} \sigma^{-(1+2r/\alpha)} \mathcal{G}(\sigma) \left[f(\sigma)^{p^+-1} + f(\sigma)^{p^--1} \right] d\sigma < \infty \quad (9)$$

is satisfied.

Remark 1.11: Here are some comments about Theorem 1.10.

- (a) The solution given in Theorem 1.3 (i) for $u_0 \in L^r(\Omega)$ is in the class defined in (8) with $\alpha = N$.
- (b) The solution given by Theorem 1.6 (i) for $u_0 \in \mathcal{I}^p(x_0) \subset L^r(\Omega)$ is in the class defined in (8) with $\alpha = \rho$.
- (c) When $f(t) = t$, $t \geq 0$, we have $\mathcal{G}(t) = 1$, $t > 0$. Then, condition (9) is verified for $p^+ < 1 + 2r/\alpha$. In particular, the solution obtained in Corollary 1.5 (i) is unique in the class defined by (8), with $\alpha = N$, because $p^+ < p^* = 1 + 2r/N$. On the other hand, the solution of Corollary 1.8 (i) is unique in the class defined by (8), with $\alpha = \rho$, since $p_0 < 1 + 2r/\rho$ and $p_0 \geq p^+$.

The remainder of the paper is organized as follows. In Section 2 we give some preliminary results about the heat semigroup, Jensen's inequality with $p(x)$ and auxiliary problems. Theorems 1.3 and 1.6 are shown in Section 3. The uniqueness is proved in Section 4.

2. Preliminaries

We denote by $P_\Omega(x, y; t)$ the Dirichlet heat kernel associated to the operator $\partial_t - \Delta_\Omega$, where $-\Delta_\Omega$ denotes the Dirichlet Laplacian operator on $\Omega \subset \mathbb{R}^N$, that is

$$S(t)u_0(x) = \int_\Omega P_\Omega(x, y; t)u_0(y) dy, \quad (10)$$

for $u_0 \in L^1(\Omega)$ and $\{S(t)\}_{t \geq 0}$ the heat semigroup. It is very well known that

$$P_D(x, y; t) \leq P_E(x, y; t) \leq P_N(x, y; t) \quad (11)$$

for $x, y \in D \subset E$, where D and E are open subsets of $\mathbb{R}^N$, and P_N is the heat kernel defined by

$$P_N(x, y; t) := P_{\mathbb{R}^N}(x, y; t) = (4\pi t)^{-N/2} \exp(-|x - y|^2/4t).$$

The following lemma is used in the proof of the non-existence results. Item (i) was established in [32], and item (ii) in [35] for $x_0 = 0$, but the methods used by them can be adapted to the case $x_0 \neq 0$.

Lemma 2.1: Let $l, \delta > 0$ such that $B_{l+2\delta}(x_0) \subset \Omega$, and $0 < \gamma < N$. There exist constants $c, c' > 0$, depending on N, x_0 and γ , such that

- (i) $S(t)\chi_{B_l(x_0)} \geq c l^N (l + \sqrt{t})^{-N} \chi_{B_{l+\sqrt{t}}(x_0)}$, for all $0 < t \leq \delta^2$.
- (ii) $S(t)|\cdot - x_0|^{-\gamma} \chi_{B_l(x_0)} \geq c' t^{-(\gamma/2)} \chi_{B_{\sqrt{t}}(x_0)}$, for all $0 < t \leq \min\{\delta^2, l^2\}$.

The notion of L^r -solution that we use in the work is the following.

Definition 2.2: Let $u_0 \in L^r(\Omega)$, $u_0 \geq 0$, $1 \leq r < \infty$, $p \in C(\Omega, \mathbb{R}_+)$ verifying (2) and $f \in C([0, +\infty))$. A measurable non-negative function u defined a.e in $\Omega \times (0, T)$ is called a L^r -solution (resp. supersolution) of problem (1) if $u \in L^\infty((0, T), L^r(\Omega))$ and

$$u(t) = \mathfrak{F}(u, u_0)(t) \text{ (resp. } u(t) \geq \mathfrak{F}(u, u_0)(t)) \quad (12)$$

a.e. in $\Omega \times (0, T)$, where

$$\mathfrak{F}(u, u_0)(t) = S(t)u_0 + \int_0^t S(t-\sigma)f(u(\sigma))^{p(\cdot)} d\sigma.$$

The following result is an adapted version of [36, Theorem 1].

Lemma 2.3: Assume that $f \in C([0, +\infty))$ is non-decreasing, $p \in C(\Omega, \mathbb{R}_+)$ verifying (2), and $u_0 \in L^r(\Omega)$, $u_0 \geq 0$ with $1 \leq r < \infty$. Then, problem (1) admits a L^r -solution in $\Omega \times (0, T)$ if and only if it admits a supersolution in $\Omega \times (0, T)$.

Proof: By Definition 2.2, any L^r -solution of (1) is a supersolution of (1). Thus, we only need to show that the existence of a supersolution $\bar{u}$ for (1) implies the existence of a L^r -solution u for (1).

Since $\bar{u}$ is supersolution of (1) we have that $\bar{u} \geq \mathfrak{F}(\bar{u}, u_0)$. Moreover, $\mathfrak{F}(\cdot, u_0)$ is non-decreasing, because f is non-decreasing and the heat semigroup $\{S(t)\}_{t \geq 0}$ is monotone. Consider now the sequence $\{u^n\}_{n \geq 0}$, given by $u^0 = \bar{u}$, and $\mathfrak{F}(u^{n-1}, u_0) = u^n$ for $n \geq 1$. Then, the sequence $\{u^n\}_{n \geq 0}$ non-increasing a.e in $\Omega \times (0, T)$ and $\bar{u} \geq u^n \geq 0$. Let $u(x, t) = \lim_{n \rightarrow +\infty} u^n(x, t)$, whenever it exists. Then from the continuity of the non-decreasing function f , the monotonicity of $\{S(t)\}_{t \geq 0}$, and by application of the monotone convergence theorem we obtain that $u = \lim_{n \rightarrow +\infty} u^n = \lim_{n \rightarrow +\infty} \mathfrak{F}(u^{n-1}, u_0) = \mathfrak{F}(u, u_0)$. Due to the construction of the sequence it is possible to conclude that $0 \leq u \leq \bar{u}$. Moreover, since $\bar{u} \in L^\infty((0, T), L^r(\Omega))$ we conclude that $u \in L^\infty((0, T), L^r(\Omega))$ and the proof is complete. ■

We recall the well-known smoothing effect of the heat semigroup on Lebesgue spaces, that is

$$\|S(t)\psi\|_{L^{q_2}} \leq (4\pi t)^{-(N/2)(1/q_1-1/q_2)} \|\psi\|_{L^{q_1}}, \quad (13)$$

for $1 \leq q_1 \leq q_2 \leq \infty$, $t > 0$ and $\psi \in L^{q_1}(\Omega)$ (see [22, Lemma 7]). We also use the following result, see [37, Proposition 2.1].

Lemma 2.4: Let $\gamma \in (0, N)$, and let $q_1, q_2 \in (1, +\infty]$ satisfy

$$0 \leq \frac{1}{q_2} < \frac{\gamma}{N} + \frac{1}{q_1} < 1.$$

There exists a constant $C_0 > 0$, depending on N, γ, q_1 and q_2 , such that

$$\|S(t)(|\cdot|^{-\gamma} \psi)\|_{L^{q_2}} \leq C_0 t^{-(N/2)(1/q_1-1/q_2)-\frac{\gamma}{2}} \|\psi\|_{L^{q_1}},$$

for all $t > 0$ and all $\psi \in L^{q_1}(\Omega)$.

The following version of the Jensen's inequality was established in [20, Lemma 3.1].

Lemma 2.5: Let η be a positive measure in $\Omega \subset \mathbb{R}^N$ such that $\int_{\Omega} d\eta = 1$. Assume that $f \in L^{p^+}(\Omega, d\eta)$, $p \in C(\Omega, \mathbb{R}_+)$ verifying (2) with $p^- \geq 1$. Then

$$\int_{\Omega} |f(x)|^{p(x)} d\eta(x) \geq 2^{-p^+} \min \left\{ \left(\int_{\Omega} |f(x)| d\eta(x) \right)^{p^-}, \left(\int_{\Omega} |f(x)| d\eta(x) \right)^{p^+} \right\}.$$

The following proposition is used in the proof of our non-existence results. Its proof follows the same ideas of [33], [16, p. 9] and [38, Lemma 15.6].

Proposition 2.6: Let $p \in C(\Omega, \mathbb{R}_+)$ with $p^- \geq 1$. Assume that $f \in C([0, +\infty))$ is non-decreasing, convex, $f(0) = 0$ and $f(s) > 0$ for all $s \geq z_0$, for some $z_0 > 0$, such that

$$\int_{z_0}^{+\infty} \frac{d\sigma}{\min \{f(\sigma)^{p^+}, f(\sigma)^{p^-}\}} < \infty. \quad (14)$$

Suppose that $u_0 : \Omega \rightarrow [0, +\infty)$, $u_0 \not\equiv 0$ and $u : \Omega \times [0, T] \rightarrow [0, +\infty)$ are measurable functions satisfy

$$S(t)u_0 + \int_0^t S(t-\sigma)f(u(\sigma))^{p(\cdot)} d\sigma \leq u(t) < \infty \quad \text{a.e. in } \Omega \times (0, T).$$

Then

$$2^{-p^+} t \cdot \left(\int_{\|S(t)u_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{\min \{f(\sigma)^{p^+}, f(\sigma)^{p^-}\}} \right)^{-1} \leq 1$$

for all $t \in (0, T]$.

Proof: Using the convexity of f , $f(0) = 0$, Jensen's inequality, $0 < \eta = \int_{\Omega} P_{\Omega}(x, y; t) dy \leq 1$ (see (11)), (10) we have

$$\begin{aligned} f(S(t)u) &= f \left(\int_{\Omega} P_{\Omega}(x, y; t) u(y) dy \right) \\ &= f \left(\eta \int_{\Omega} \frac{P_{\Omega}(x, y; t)}{\eta} u(y) dy \right) \\ &\leq \eta f \left(\int_{\Omega} \frac{P_{\Omega}(x, y; t)}{\eta} u(y) dy \right) \\ &\leq \int_{\Omega} P_{\Omega}(x, y; t) f(u(y)) dy = S(t)f(u) \end{aligned} \quad (15)$$

for all $t > 0$.

On the other hand, since

$$u(t) \geq S(t)u_0 + \int_0^t S(t-\sigma)f(u(\sigma))^{p(\cdot)} d\sigma,$$

for $t \in (0, T)$, taking $0 < t < s < T$ we have

$$S(s-t)u(t) \geq S(s)u_0 + \int_0^t S(s-\sigma)f(u(\sigma))^{p(\cdot)} d\sigma. \quad (16)$$

Set

$$\Theta(\cdot, t) := S(s)u_0 + \int_0^t S(s-\sigma)f(u(\sigma))^{p(\cdot)} d\sigma$$

for all $(x, t) \in \Omega \times (0, s)$.

Note that for $x \in \Omega$ fixed the function $\Theta(x, t)$ is absolutely continuous on $(0, s)$, consequently it is differentiable a.e. in $(0, s)$ and from (16) we have

$$\Theta'(t) = S(s-t)f(u(t))^{p(\cdot)} \quad \text{a.e. } t \in (0, s),$$

and from Lemma 2.5

$$\begin{aligned} S(s-t)f(u(t))^{p(\cdot)} &= \int_{\Omega} P_{\Omega}(x, y; s-t)f(u(t, y))^{p(y)} dy \\ &\geq 2^{-p^+} \min \left\{ \frac{[S(s-t)f(u(t))]^{p^-}}{a(s-t, x)^{p^- - 1}}, \frac{[S(s-t)f(u(t))]^{p^+}}{a(s-t, x)^{p^+ - 1}} \right\}, \end{aligned}$$

where P_{Ω} is the Dirichlet heat kernel on Ω and $a(s-t, x) = \int_{\Omega} P_{\Omega}(x, y; s-t) dy > 0$. Since $P_{\Omega}(x, y; s-t) \leq P_{\mathbb{R}^N}(x, y; s-t)$ (see [39, Lemma 7]), we conclude that $a(s-t, x) \leq 1$. Thus, since $p^- \geq 1, f$ is non-decreasing, inequality (15) and (16) we obtain

$$\begin{aligned} \Theta'(t) &\geq 2^{-p^+} \min \left\{ [S(s-t)f(u(t))]^{p^-}, [S(s-t)f(u(t))]^{p^+} \right\} \\ &\geq 2^{-p^+} \min \left\{ [f(S(s-t)u(t))]^{p^+}, [f(S(s-t)u(t))]^{p^-} \right\} \\ &\geq 2^{-p^+} \min \left\{ [f(\Theta(t))]^{p^+}, [f(\Theta(t))]^{p^-} \right\}. \end{aligned} \tag{17}$$

Set $g(t) = \min\{f(t)^{p^-}, f(t)^{p^+}\}$ for all $t \geq 0$. By (17), $\Theta'(t) \geq 2^{-p^+}g(\Theta(t))$. Defining $H(t) = \int_t^{+\infty} d\sigma/g(\sigma)$ for $t > 0$ we obtain $[H(\Theta(t))] = -(\Theta'(t)/g(\Theta(t))) \leq -2^{-p^+}$, for $0 < t < s$. Note that condition (14) guarantees that the function H is well defined.

Integrating from 0 to s we get $H(\Theta(s)) - H(\Theta(0)) \leq -2^{-p^+}s$, that is

$$2^{-p^+}s \leq \int_{\Theta(0)}^{+\infty} \frac{d\sigma}{g(\sigma)} - \int_{\Theta(s)}^{+\infty} \frac{d\sigma}{g(\sigma)} \leq \int_{\Theta(0)}^{+\infty} \frac{d\sigma}{g(\sigma)} = \int_{S(s)u_0}^{+\infty} \frac{d\sigma}{g(\sigma)}.$$

Thus, since H is decreasing and the left hand does not depend on x , we conclude that

$$2^{-p^+}s \leq \int_{\|S(s)u_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{g(\sigma)}$$

and the result follows. ■

The non-existence part of L^1 -solutions will be shown using the following result, see [32, Lemma 4.2].

Lemma 2.7: *Suppose that $f : [0, +\infty) \rightarrow [0, +\infty)$ is non-decreasing and $q > 1, p^- > 0$. Then the following two conditions are equivalent*

- (i) *There exists a sequence $\{s_k\}_{k \in \mathbb{N}}$ such that $s_{k+1} \geq \theta s_k$, $\theta > 1$ and $\sum_{k=1}^{+\infty} s_k^{-q} f(s_k)^{p^-} = +\infty$.*
- (ii) *$\int_1^{+\infty} \sigma^{-q} F(\sigma) d\sigma = +\infty$, where $F(s) = \sup_{1 \leq t \leq s} f(t)^{p^-}/t$.*

Laister et al. [32] showed the existence of a L^r -solution for problem (3). In the following result, we establish the continuity of this solution.

Proposition 2.8: *Let $f \in C([0, +\infty))$ be non-decreasing, $u_0 \in L^r(\Omega)$, $u_0 \geq 0$ with $1 \leq r < \infty$, and $p^* = 1 + 2r/N$. Assume that*

$$\begin{aligned} \limsup_{t \rightarrow +\infty} t^{-p^*} f(t) &< \infty \quad \text{if } r > 1, \\ \int_1^{+\infty} \sigma^{-p^*} F(\sigma) \, d\sigma &< \infty \quad \text{if } r = 1, \end{aligned}$$

where $F(\sigma) = \sup_{1 \leq t \leq \sigma} f(t)/t$ and $u \in L^\infty((0, T), L^r(\Omega))$ is the L^r -solution of problem (3) obtained in [32]. Then, u belongs to the set

$$\mathcal{X} = C([0, T], L^r(\Omega)) \cap \{v \in L_{loc}^\infty((0, T), L^\infty(\Omega)); \sup_{t \in (0, T]} \|v(t)\|_{L^\infty} < \infty\}. \quad (18)$$

Proof: Since $u_0 \in L^r(\Omega)$, $w_1 = S(\cdot)u_0 \in C([0, T], L^r(\Omega))$. From this and estimate (13), it follows that $w_1 \in \mathcal{X}$. Thus, it is sufficient to show that $w_2 = u(\cdot) - w_1$ also belongs to $\mathcal{X}$, see (12) with $p(x) = 1$ constant.

Assume first $r > 1$. Due to $\limsup_{t \rightarrow +\infty} t^{-p^*} f(t) < \infty$, there exists $\kappa > 0$ such that

$$f(t) \leq \kappa(1 + t^{p^*}) \quad (19)$$

for all $t \geq 0$. In the proof of Theorem 1.1 the authors used, as a supersolution for problem (3), the solution of the problem

$$\begin{cases} u_t - \Delta u = \kappa(1 + u^{p^*}) & \text{in } \Omega \times (0, T) \\ u = 0 & \text{on } \partial\Omega \times (0, T) \\ u(0) = u_0 \geq 0 & \text{in } \Omega. \end{cases} \quad (20)$$

From [22, p. 284], the solution w of problem (20) belongs to the set

$$E = L^\infty((0, T), L^r(\Omega)) \cap \{u \in L_{loc}^\infty((0, T), L^\eta(\Omega)); t^\beta u \in C_0([0, T], L^\eta(\Omega))\},$$

with $\eta \in (r, p^*r)$, $\eta \geq p^*$ and $\beta = (N/2)(1/r - 1/\eta) < 1/p^* < 1$. Here, C_0 means the functions that vanish at $t = 0$. Moreover, from Theorem 1 of [22], w is non-negative and $t^{N/2r} \|w(t)\|_{L^\infty} \leq C$ for $t \in (0, T]$ and some constant $C > 0$.

From Lemma 2.3, with $p(x) = 1$ constant, we have that

$$0 \leq u(t) \leq w(t). \quad (21)$$

To show that $w_2 \in \mathcal{X}$ we argue as follows. From (21) we obtain $t^{N/2r} \|w_2(t)\| \leq C$ for $t \in (0, T]$ and some constant $C > 0$. The continuity of w_2 is proved as [27, p. 285]. Let $0 \leq \tau < t < T$. Using (19), (21) and the fact that f is non-decreasing, we obtain

$$\begin{aligned} \left\| \int_\tau^t S(t-\sigma) f(u(\sigma)) \, d\sigma \right\|_{L^r} &\leq \left\| \int_\tau^t S(t-\sigma) f(w(\sigma)) \, d\sigma \right\|_{L^r} \\ &\leq \kappa \left\| \int_\tau^t S(t-\sigma) (1 + w^{p^*}) \, d\sigma \right\|_{L^r} \\ &\leq C(t-\tau) + \kappa \left\| \int_\tau^t S(t-\sigma) w^{p^*}(\sigma) \, d\sigma \right\|_{L^r}, \end{aligned}$$

and

$$\begin{aligned}
 \left\| \int_{\tau}^t S(t-\sigma) w^{p^*}(\sigma) d\sigma \right\|_{L^r} &\leq \int_{\tau}^t (t-\sigma)^{-(N/2)(p^*/\eta-1/r)} \|w^{p^*}(\sigma)\|_{L^{\eta/p^*}} d\sigma \\
 &\leq \left(\sup_{\sigma \in (\tau, t)} \sigma^{\beta} \|w(\sigma)\|_{L^{\eta}} \right)^{p^*} \int_{\tau}^t (t-\sigma)^{-(N/2)(p/\eta-1/r)} \sigma^{-\beta p^*} d\sigma \\
 &= \left(\sup_{\sigma \in (\tau, t)} \sigma^{\beta} \|w(\sigma)\|_{L^{\eta}} \right)^{p^*} \int_{\tau/t}^1 (1-\sigma)^{-(N/2)(p/\eta-1/r)} \sigma^{-\beta p^*} d\sigma.
 \end{aligned}$$

From these inequalities we conclude that

$$\left\| \int_{\tau}^t S(t-\sigma) f(u(\sigma)) d\sigma \right\|_{L^r} \rightarrow 0$$

as $t \rightarrow \tau$. Indeed, this is clear if $\tau > 0$, and when $\tau = 0$ we use that $\sup_{\sigma \in (0, t)} \sigma^{\beta} \|w(\sigma)\|_{L^{\eta}} \rightarrow 0$ as $t \rightarrow 0$, since $w \in E$.

For $r = 1$, the supersolution used in [32] is given by $w(t) = AS(t)u_0 + \chi_{\Omega}$ with $A > 1$. Note that $\|w(t)\|_{L^{\infty}} \leq At^{-N/2} \|u_0\|_{L^1} + 1 \leq C_1 t^{-N/2}$ for some constant $C_1 > 0$. Thus, as the previous case, we obtain $t^{N/2} \|w_2(t)\|_{L^{\infty}} \leq C$ for $t \in (0, T]$ and some constant $C > 0$.

The continuity of w_2 follows from the fact: if $g \in L^1((0, T), X)$, X is a Banach space and $v(t) = \int_0^t S(t-\sigma)g(\sigma) d\sigma$, then $v \in C([0, T], X)$.

Due to (21) and the non-decreasing property of f we have

$$\begin{aligned}
 \int_0^T \|f(u(\sigma))\|_{L^1} d\sigma &\leq \int_0^T \|f(w(\sigma))\|_{L^1} d\sigma \\
 &\leq \int_0^T F(\|w(\sigma)\|_{L^{\infty}}) \|w(\sigma)\|_{L^1} d\sigma \\
 &\leq C \int_0^T F(C_1 \sigma^{-N/2}) d\sigma \\
 &\leq C \int_{CT^{-N/2}}^{\infty} F(\sigma) \sigma^{-(1+2/N)} d\sigma < \infty.
 \end{aligned}$$

Therefore, $f(u(\cdot)) \in L^1((0, T), L^1(\Omega))$. ■

3. Existence and non-existence

Proof of Theorem 1.3.: Set $g, h \in C([0, \infty))$ non-decreasing functions defined by $h(t) = \min\{f(t)^{p^-}, f(t)^{p^+}\}$ and $g(t) = f(t)^{p^+} + f(t)^{p^-}$ for all $t \geq 0$. Note that for all $x \in \Omega$,

$$h(t) \leq f(t)^{p(x)} \leq g(t), \quad (22)$$

for all $t \geq 0$ and $0 < p^- \leq p(x) \leq p^+$.

(i) Due to (5) and (6) we have

$$\begin{aligned}
 \limsup_{t \rightarrow +\infty} t^{-p^*} g(t) &< \infty \quad \text{for } r > 1, \\
 \int_1^{+\infty} \sigma^{-p^*} G(\sigma) d\sigma &< \infty \quad \text{for } r = 1,
 \end{aligned}$$

where $G(\sigma) = \sup_{1 \leq t \leq \sigma} g(t)/t$. By Proposition 2.8 the problem (3), with g in place of f , has a L^r -solution $v \in L^\infty((0, T), L^r(\Omega))$ on some interval $(0, T)$. From (22), v is a supersolution of (1) and the existence of a L^r -solution u defined on $(0, T)$ follows from Lemma 2.3. Moreover, as $0 \leq u \leq v$ in $[0, T)$ we can argue as in the proof of the Proposition 2.8 to conclude that $u \in \mathcal{X}$, defined by (18).

- (ii) Without loss of generality we assume that $0 \in \Omega$. The proof follows similar arguments as those used in [35] and [32]. To do this, one considers an initial datum $u_0(x) = \sum_{i=1}^{+\infty} u_n(x)$ with $u_n(x) = c_0 z_n \chi_{\sigma_n}(x)$, where $\sigma_n = (z_n n^2)^{-r/N}$, c_0 is a suitable constant, and $\lim_{n \rightarrow +\infty} z_n = +\infty$. Note that $u_0 \in L^r(\Omega)$,

$$\|u_0\|_{L^r} \leq c_0 \omega_N^{1/r} \sum_{n=1}^{+\infty} \frac{1}{n^2} < \infty,$$

and

$$\lim_{n \rightarrow +\infty} u_n(x) = \begin{cases} +\infty & \text{if } x = 0, \\ 0 & \text{if } x \neq 0. \end{cases}$$

We analyze two situations:

Case $r > 1$. Assume $\limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^-} = +\infty$. Then, there exists a sequence $\{\phi_n\}_{n \geq 1}$ such that $\phi_n \geq n$ and $f(\phi_n)^{p^-} \geq e^{n/r} \phi_n^{p^*} \geq 1$.

Let $\rho_n = (\phi_n n^2)^{-r/N}$. Thus, since $\lim_{n \rightarrow +\infty} \rho_n = 0$, it is possible to assume that $B_{3\rho_n} \subset \Omega$ for every $n \geq 1$. Let $u_0 = \sum_{n=1}^{+\infty} u_n$, where $u_n = c^{-1} 2^N \phi_n \chi_{\rho_n}$, and c is the constant given in Lemma 2.1 (i). Note that

$$\|u_0\|_{L^r} \leq \sum_{n=1}^{+\infty} c^{-1} 2^N \phi_n \omega_N^{1/r} \rho_n^{N/r} = c^{-1} 2^N \omega_N^{1/r} \sum_{n=1}^{+\infty} n^{-2} < \infty,$$

where ω_N is the volume of the unit ball. Now, we argue by contradiction and assume that $u \in L^\infty((0, T), L^r(\Omega))$ is a L^r -solution of problem (1) with $u(0) = u_0$, for some $T > 0$. Since $u \geq 0$ and $f \geq 0$, by Lemma 2.1 (i) we have

$$u(\sigma) \geq S(\sigma)u_0 \geq S(\sigma)u_n \geq 2^N \phi_n \left(\frac{\rho_n}{\rho_n + \sqrt{\sigma}} \right)^N \chi_{\rho_n + \sqrt{\sigma}} \geq \phi_n \chi_{\rho_n},$$

for $0 < \sigma \leq \rho_n^2$. Since f is non-decreasing we have

$$f(u(\sigma)) \geq f(\phi_n) \chi_{\rho_n}, \quad 0 < \sigma \leq \rho_n^2. \quad (23)$$

Thus, for $t \in [\rho_n^2/2, \rho_n^2]$, by (12), (22), (23) and Lemma 2.1 (i), we have

$$\begin{aligned} u(t) &= S(t)u_0 + \int_0^t S(t-\sigma) f(u(\sigma))^{p(\cdot)} d\sigma \\ &\geq S(t)u_0 + \int_0^t S(t-\sigma) [f(\phi_n) \chi_{\rho_n}]^{p(\cdot)} d\sigma \\ &\geq \int_0^t S(t-\sigma) h(\phi_n) \chi_{\rho_n} d\sigma \\ &= \int_0^t S(t-\sigma) [f(\phi_n)]^{p^-} \chi_{\rho_n} d\sigma \end{aligned}$$

$$\begin{aligned}
&\geq f(\phi_n)^{p^-} \int_0^t S(t-\sigma) \chi_{\rho_n} d\sigma \\
&\geq Cf(\phi_n)^{p^-} \int_0^t \chi_{\rho_n + \sqrt{t-\sigma}} d\sigma \\
&\geq Cf(\phi_n)^{p^-} \int_0^{\rho_n^2/2} \chi_{\rho_n} d\sigma \\
&\geq Cf(\phi_n)^{p^-} \rho_n^2 \chi_{\rho_n}.
\end{aligned}$$

Hence,

$$\begin{aligned}
\|u(t)\|_{L^r} &\geq Cf(\phi_n)^{p^-} \rho_n^2 \|\chi_{\rho_n}\|_{L^r} \\
&= C\omega_N^{1/r} f(\phi_n)^{p^-} \rho_n^{2+N/r} \\
&\geq C\omega_N^{1/r} e^{n/r} \phi_n^{p^*} (\phi_n^{-1} n^{-2})^{p^*} \\
&= \omega_N^{1/r} C e^{n/r} n^{-2p^*}.
\end{aligned}$$

Since the right-hand side tends to infinity as $n \rightarrow +\infty$, we conclude that $u \notin L^\infty((0, T), L^r(\Omega))$, and we have a contradiction.

Case $r = 1$: Assume that $\int_1^{+\infty} \sigma^{-p^*} F_2(\sigma) d\sigma = +\infty$. By Lemma 2.7, there exists a sequence $\{s_k\}_{k \geq 0}$ such that $s_{k+1} \geq \theta s_k$, for some $\theta > 1$, and $\sum_{k=1}^{+\infty} s_k^{-(1+2/N)} f(s_k)^{p^-} = +\infty$. Let $\delta > 0$ be small enough so that $B_{3\delta} \subset \Omega$ and consider a sequence $\{\psi_n\}_{n \geq 1}$, with $\psi_n^N = n^2 s_{k_n+1} > 0$ (the sequence $\{k_n\}_{n \geq 1}$ will be chosen later) such that $1/\psi_n < \delta$ for $n \geq n_0$, for some $n_0 \in \mathbb{N}$.

Now, we argue by contradiction and assume that for every $u_0 \in L^1(\Omega)$ there exists a L^1 -solution of problem (1). Set $u_0 = 2^N c^{-1} \sum_{n=1}^{+\infty} n^{-2} \psi_n^N \chi_{1/\psi_n}$, where c is the constant of Lemma 2.1 (i). Note that

$$\|u_0\|_{L^1} \leq 2^N c^{-1} \omega_N \sum_{n=1}^{+\infty} n^{-2} < \infty.$$

Thus, by the assumption, there exists $u \in L^\infty((0, T), L^1(\Omega))$ a L^1 -solution of problem (1) with $u(0) = u_0$, for some $T > 0$. By Lemma 2.1 (i)

$$S(\sigma)u_0 \geq 2^N c^{-1} n^{-2} \psi_n^N S(\sigma) \chi_{1/\psi_n} \geq n^{-2} \sigma^{-N/2} \chi_{1/\psi_n + \sqrt{\sigma}},$$

for $1/\psi_n^2 \leq \sigma \leq \delta^2$ and $n \geq n_0$. Since f is non-decreasing and (22) we have

$$\begin{aligned}
u(t) &\geq \int_0^t S(t-\sigma) f(u(\sigma))^{p(\cdot)} d\sigma \\
&\geq \int_0^t S(t-\sigma) h(u(\sigma)) d\sigma \\
&\geq \int_0^t S(t-\sigma) h(S(\sigma)u_0) d\sigma \\
&\geq \int_{1/\psi_n^2}^t S(t-\sigma) h(n^{-2} \sigma^{-N/2} \chi_{1/\psi_n + \sqrt{\sigma}}) d\sigma \\
&\geq \int_{1/\psi_n^2}^t h(n^{-2} \sigma^{-N/2}) S(t-\sigma) \chi_{1/\psi_n + \sqrt{\sigma}} d\sigma
\end{aligned}$$

for $1/\psi_n^2 \leq \sigma \leq t < \delta^2$. From Lemma 2.1 (i), the estimates $\sqrt{\sigma} + \sqrt{t-\sigma} \geq \sqrt{t}$ for all $0 \leq \sigma \leq t$, and $1/\psi_n + \sqrt{\sigma} + \sqrt{t-\sigma} \leq 3\sqrt{t}$ for all $1/\psi_n^2 \leq \sigma \leq t$, we obtain

$$\begin{aligned} u(t) &\geq c \int_{1/\psi_n^2}^t h(n^{-2}\sigma^{-N/2}) \left(\frac{1/\psi_n + \sqrt{\sigma}}{1/\psi_n + \sqrt{\sigma} + \sqrt{t-\sigma}} \right)^N \chi_{1/\psi_n + \sqrt{\sigma} + \sqrt{t-\sigma}} d\sigma \\ &\geq Ct^{-N/2} \chi_{1/\psi_n + \sqrt{t}} \int_{1/\psi_n^2}^t h(n^{-2}\sigma^{-N/2}) \sigma^{N/2} d\sigma. \end{aligned}$$

Thus, making a change in the variable we have

$$\begin{aligned} \|u(t)\|_{L^1} &\geq C \int_{1/\psi_n^2}^t h(n^{-2}\sigma^{-N/2}) \sigma^{N/2} d\sigma \\ &\geq C n^{-(2N+4)/N} \int_{t^{-N/2}}^{s_{k_n}+1} h(\sigma) \sigma^{-2-2/N} d\sigma, \end{aligned} \quad (24)$$

for $n \geq n_0$.

Note that there exists $\sigma_0 > 1$ such that $f(\sigma_0) > 1$. Indeed, assume by contradiction that $f(\sigma) \leq 1$ for all $\sigma > 1$, then

$$+\infty = \int_1^{+\infty} \sigma^{-p^*} F_2(\sigma) d\sigma \leq \int_1^{+\infty} \sigma^{-p^*} \sup_{1 \leq t \leq \sigma} \frac{f(t)^{p^-}}{t} d\sigma \leq \int_1^{+\infty} \sigma^{-p^*} d\sigma < \infty,$$

which yields a contradiction. Hence, since f is non-decreasing, $f(\sigma) > 1$ for $\sigma \geq \sigma_0$, and $h(\sigma) = f(\sigma)^{p^-}$ for $\sigma \geq \sigma_0$.

Fix $t \in (0, \min\{T, \sigma_0^{-2/N}\})$ and let $n_1 \geq n_0$ such that $s_{k_{n_1}} \geq t^{-N/2}$, from (24) we have

$$\begin{aligned} \|u(t)\|_{L^1} &\geq C n^{-(2N+4)/N} \int_{t^{-N/2}}^{s_{k_n}+1} f(\sigma)^{p^-} \sigma^{-2-2/N} d\sigma \\ &\geq C n^{-(2N+4)/N} \sum_{j=k_{n_1}}^{k_n} \int_{s_j}^{s_{j+1}} f(\sigma)^{p^-} \sigma^{-2-2/N} d\sigma \\ &\geq C n^{-(2N+4)/N} \sum_{j=k_{n_1}}^{k_n} f(s_j)^{p^-} \int_{s_j}^{s_{j+1}} \sigma^{-2-2/N} d\sigma \\ &= C n^{-(2N+4)/N} \sum_{j=k_{n_1}}^{k_n} f(s_j)^{p^-} \left[\frac{s_{j+1}^{-1-2/N} - s_j^{-1-2/N}}{-1-2/N} \right] \\ &= C n^{-(2N+4)/N} \sum_{j=k_{n_1}}^{k_n} f(s_j)^{p^-} s_j^{-1-2/N} \left[1 - \left(\frac{s_{j+1}}{s_j} \right)^{-1-2/N} \right] \\ &\geq (1 - \theta^{-1-2/N}) C n^{-(2N+4)/N} \sum_{j=k_{n_1}}^{k_n} f(s_j)^{p^-} s_j^{-(1+2/N)}, \end{aligned} \quad (25)$$

for all $n > n_1$. Hence, since that $\sum_{k=1}^{+\infty} s_k^{-(1+2/N)} f(s_k)^{p^-} = +\infty$ there exists a subsequence $\{k_n\}_{n \geq 1}$ such that $n^{-(2N+4)/N} \sum_{j=k_{n_1}}^{k_n} f(s_j)^{p^-} s_j^{-(1+2/N)}$ diverges as $n \rightarrow +\infty$. From this and the estimate (25) we have that $u \notin L^\infty((0, T), L^1(\Omega))$. This is a contradiction. ■

Proof of Theorem 1.6.: (i) We adapt the ideas used in [33]. Since f is convex, non-decreasing, $p^- \geq 1$, and $f(0) = 0$ the function $G : (0, +\infty) \rightarrow [0, +\infty)$ given by $G(t) = [f(t)^{p^+} + f(t)^{p^-}]/t$ is well defined, continuous and non-decreasing.

Using the assumptions $\limsup_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^-} < \infty$ and $\limsup_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^+} < \infty$ for $p_0 \geq p^*$, there exists $t_0 > 1$ so that $f(t)^{p^+} + f(t)^{p^-} \leq \eta t^{p_0}$ for some $\eta > 0$ and all $t \geq t_0$. Thus,

$$\int_1^{+\infty} \sigma^{-(1+\frac{2r}{\rho})} G(\sigma) d\sigma \leq C + \eta \int_{t_0}^{+\infty} \sigma^{-2-\frac{2r}{\rho}+p_0} d\sigma < \infty, \quad (26)$$

because $0 < \rho < 2r/(p_0 - 1) \leq N$.

Without loss of generality we assume that $x_0 = 0 \in \Omega$. Let $w(t) = AS(t)u_0$ with $A > 1$, $u_0 \in \mathcal{I}^\rho$, and $\rho < 2r/(p_0 - 1)$. Thus, there exist $K > 0$ and $a > 0$ such that $u_0 \leq K^{1/r} \cdot |\cdot|^{-\rho/r} \chi_{B_a}$. If $u_0 = 0$ then $u = 0$ is a L^r -solution since $f(0) = 0$. Assume now that $u_0 \neq 0$. From Lemma 2.4, we have $\|w(t)\|_{L^\infty} \leq AC_0 K^{1/r} t^{-\rho/2r}$, for some constant $C_0 > 0$ and for all $t > 0$. From (22) and the bound (26)

$$\begin{aligned} S(t)u_0 &+ \int_0^t S(t-\sigma) f(w(\sigma))^{p(\cdot)} d\sigma \\ &\leq S(t)u_0 + \int_0^t S(t-\sigma) [f(w(\sigma))^{p^+} + f(w(\sigma))^{p^-}] d\sigma \\ &= S(t)u_0 + \int_0^t S(t-\sigma) G(w(\sigma)) w(\sigma) d\sigma \\ &\leq S(t)u_0 + A \int_0^t G(\|w(\sigma)\|_{L^\infty}) S(t-\sigma) S(\sigma) u_0 d\sigma \\ &= S(t)u_0 + w(t) \int_0^t G(AC_0 K^{1/r} \sigma^{-\rho/2r}) d\sigma \\ &= S(t)u_0 + w(t) (AC_0 K^{1/r})^{2r/\rho} \left(\frac{2r}{\rho}\right) \int_{t^{-\rho/2r} AC_0 K^{1/r}}^{+\infty} \sigma^{-(1+2r/\rho)} G(\sigma) d\sigma \\ &= \left\{ \frac{1}{A} + (AC_0 K^{1/r})^{2r/\rho} \left(\frac{2r}{\rho}\right) \int_{t^{-\rho/2r} AC_0 K^{1/r}}^{+\infty} \sigma^{-(1+2r/\rho)} G(\sigma) d\sigma \right\} w(t) \\ &\leq w(t), \end{aligned} \quad (27)$$

for $t \in (0, T)$ with $T > 0$ sufficiently small. Thus, $\mathfrak{F}(w, u_0) \leq w$ in $(0, T)$ and w is a supersolution of (1) in $(0, T)$.

It follows from Lemma 2.3 that there exists a L^r -solution of problem (1) such that $0 \leq u \leq w$ in $(0, T)$. In addition, $\|u(t)\|_{L^\infty} \leq \|w(t)\|_{L^\infty} \leq Ct^{-\rho/2r}$ for $t \in (0, T)$ and some constant $C > 0$.

Finally, we show the continuity of u . Since $S(\cdot)u_0 \in C([0, T], L^r(\Omega))$, we only need to show the same for $u(\cdot) - S(\cdot)u_0$. To this end, we proceed as in the proof of Proposition 2.8. Thus,

$$\begin{aligned} \int_0^T \|f(u(\sigma))^{p(\cdot)}\|_{L^r} d\sigma &\leq \int_0^T \|f(w(\sigma))^{p(\cdot)}\|_{L^r} d\sigma \\ &\leq \int_0^T \|f(w(\sigma))^{p^+} + f(w(\sigma))^{p^-}\|_{L^r} d\sigma \\ &\leq \int_0^T G(\|w(\sigma)\|_{L^\infty}) \|w(\sigma)\|_{L^r} d\sigma \\ &\leq C \int_0^T G(AC_0 K^{1/r} \sigma^{-\rho/2r}) d\sigma \end{aligned}$$

$$\leq C \int_{T^{-\rho/2r} AC_0 K^{1/r}}^{+\infty} G(\sigma) \sigma^{-(1+2r/\rho)} d\sigma < \infty,$$

by (26). Therefore, $f(u(\cdot))^{p(\cdot)} \in L^1((0, T), L^r(\Omega))$.

(ii) Assume first that $\liminf_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^+} > 0$, $p_0 > p^*$ and $2r/(p_0 - 1) < \rho < N$. Since p is continuous, we consider $\zeta = p^+ - \delta > 1$ and the set $\Omega_1 = p^{-1}(\zeta, +\infty) \subset \Omega$, the preimage of $(\zeta, +\infty)$, for δ sufficiently small such that

$$p_0 > p^* p^+ / \zeta > p^* \quad \text{and} \quad \rho > 2r/(\zeta p_0 / p^+ - 1) > 2r/(p_0 - 1).$$

Take a $x_0 \in \Omega_1$ and consider $u_0 \in \mathcal{I}_\rho(x_0)$, that is $u_0 \geq K^{1/r} \cdot -x_0|^{-\rho/r} \chi_{B_a(x_0)} = v_0$, where $B_a(x_0) \subset \Omega_1$. Note that $v_0 \in L^r(\Omega)$ since $0 < \rho < N$.

We argue by contradiction and suppose that there exists a non-negative L^r -solution u of (1) with initial condition $u_0 \in \mathcal{I}_\rho(x_0)$, that is

$$u(t) = S(t)u_0 + \int_0^t S(t-\sigma) f(u(\sigma))^{p(\cdot)} d\sigma \quad (28)$$

for $t \in (0, T)$, and $u \in L^\infty((0, T), L^r(\Omega))$ for some $T > 0$. From the properties of heat kernel and (28) we have

$$\begin{aligned} u(t) &\geq S_{\Omega_1}(t)u_0 + \int_0^t S_{\Omega_1}(t-\sigma) f(u(\sigma))^{p(\cdot)} d\sigma \\ &\geq S_{\Omega_1}(t)v_0 + \int_0^t S_{\Omega_1}(t-\sigma) \min \left\{ f(u(\sigma))^{p^+}, f(u(\sigma))^\zeta \right\} d\sigma \end{aligned} \quad (29)$$

for $t \in (0, T)$. Note that Lemma 2.4 implies $S_{\Omega_1}(t)v_0 \in L^\infty(\Omega_1)$ for all $t > 0$. Moreover, by Lemma 2.1, we have $S_{\Omega_1}(t)v_0 \geq c' t^{-(\rho/2r)} \chi_{B_{\sqrt{t}}(x_0)}$, for all $0 < t < \min\{(a/3)^2, T\}$. Thus

$$\|S_{\Omega_1}(t)v_0\|_{L^\infty} \geq c' t^{-\rho/2r}, \quad (30)$$

for all $0 < t < \min\{(a/3)^2, T\}$.

Since $\liminf_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^+} > 0$ with $p_0 > p^* p^+ / \zeta$ there exists a constant $\eta > 0$ such that $f(t)^{p^+} \geq \eta t^{p_0} \geq 1$ for $t \geq t_0 \geq 1$ for some constant t_0 . From (30), we see that there exists τ_0 such that $\|S_{\Omega_1}(\tau)v_0\|_{L^\infty} \geq t_0$ for all $0 < \tau \leq \tau_0$. Using again (30)

$$\begin{aligned} \int_{\|S_{\Omega_1}(\tau)v_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{\min \{f(\sigma)^{p^+}, f(\sigma)^\zeta\}} &= \int_{\|S_{\Omega_1}(\tau)v_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{f(\sigma)^\zeta} \\ &\leq \eta^{-\zeta/p^+} \int_{\|S_{\Omega_1}(\tau)v_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{\sigma^{p_0\zeta/p^+}} \\ &= \frac{\eta^{-\zeta/p^+}}{(p_0\zeta/p^+ - 1)} \|S_{\Omega_1}(\tau)v_0\|_{L^\infty}^{1-p_0\zeta/p^+} \\ &\leq \frac{\eta^{-\zeta/p^+} (c')^{1-p_0\zeta/p^+}}{(p_0\zeta/p^+ - 1)} \tau^{-\rho(1-p_0\zeta/p^+)/2r}. \end{aligned}$$

Thus, from (29) and Proposition 2.6

$$1 \geq 2^{-p^+} \tau \left(\int_{\|S_{\Omega_1}(\tau)v_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{\min \{f(\sigma)^{p^+}, f(\sigma)^\zeta\}} \right)^{-1}$$

$$\geq 2^{-p^+} \eta^{\zeta/p^+} (p_0 \zeta / p^+ - 1) (c')^{(p_0 \zeta / p^+ - 1)} \tau^{1 - \rho(p_0 \zeta / p^+ - 1)/2r}$$

for all $0 < \tau \leq \min\{\tau_0, (a/3)^2, T\}$. This is a contradiction for τ sufficiently small, since $1 - \rho(p_0 \zeta / p^+ - 1)/2r < 0$.

When $\liminf_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^-} > 0$ we argue as follows. Let $u_0 \in \mathcal{I}_\rho$, that is $u_0 \geq K^{1/r} \cdot |\cdot|^{-\rho/r} \chi_{B_a} = v_0$, where $B_a \subset \Omega$ and we suppose that there exists a non-negative L^r -solution u verifying (28). As in the derivation of (29) and (30) we obtain

$$u(t) \geq S(t)v_0 + \int_0^t S(t-\sigma) \min \left\{ f(u(\sigma))^{p^+}, f(u(\sigma))^{p^-} \right\} d\sigma, \quad (31)$$

and

$$\|S(t)v_0\|_{L^\infty} \geq c'_N t^{-\rho/2r}, \quad (32)$$

for all $0 < t < \min\{(a/3)^2, T\}$.

Since $p_0 > p^*$ such that $\liminf_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^-} > 0$. Thus, there exists a constant $\eta > 0$ such that $f(t)^{p^-} \geq \eta t^{p_0} \geq 1$ for $t \geq t_0 \geq 1$ for some constant t_0 . From (32), we see that there exists τ_0 such that $\|S(\tau)v_0\|_{L^\infty} \geq t_0$ for all $0 < \tau \leq \tau_0$, and

$$\begin{aligned} \int_{\|S(\tau)v_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{\min \{f(\sigma)^{p^+}, f(\sigma)^{p^-}\}} &= \int_{\|S(\tau)v_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{f(\sigma)^{p^-}} \\ &\leq \frac{1}{\eta} \int_{\|S(\tau)v_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{\sigma^{p_0}} \\ &= \frac{1}{\eta(p_0 - 1)} \|S(\tau)v_0\|_{L^\infty}^{1-p_0} \\ &\leq \frac{(c'_N)^{1-p_0}}{\eta(p_0 - 1)} \tau^{-\rho(1-p_0)/2r}. \end{aligned}$$

Thus, from (31) and Proposition 2.6

$$\begin{aligned} 1 &\geq 2^{-p^+} \tau \left(\int_{\|S(\tau)v_0\|_{L^\infty}}^{+\infty} \frac{d\sigma}{\min \{f(\sigma)^{p^+}, f(\sigma)^{p^-}\}} \right)^{-1} \\ &\geq 2^{-p^+} \eta (p_0 - 1) (c'_N)^{(p_0-1)} \tau^{1+\rho(1-p_0)/2r} \end{aligned}$$

for all $0 < \tau \leq \min\{\tau_0, (a/3)^2, T\}$. This is a contradiction for τ sufficiently small, since $1 + \rho(1 - p_0)/2r < 0$. ■

Remark 3.1: It is possible to show the existence of solution for problem (3) without the assumption of the convexity of f , $f(0) = 0$ and $p^- \geq 1$. To do this, we assume that $f \in C([0, +\infty))$ is non-decreasing and the condition (7) is satisfied. Define $\tilde{f} \in C([0, +\infty))$ by $\tilde{f}(t) = f(t)^{p^+} + f(t)^{p^-}$ for $t \in [0, 1]$, and $\tilde{f}(t) = t[F_1(t) + F_2(t)]$ for $t \geq 1$. Then $f^{p^+} + f^{p^-} \leq \tilde{f}$ and the function $\tilde{G} : [1, +\infty) \rightarrow [0, +\infty)$ defined by $\tilde{G}(t) = \tilde{f}(t)/t$ is continuous non-decreasing. Note that $\tilde{G}(t) = \tilde{f}(t)/t \leq F_1(t) + F_2(t)$. From (7) we have

$$\int_1^{+\infty} \sigma^{-(1+2r/\rho)} \tilde{G}(\sigma) d\sigma \leq \int_1^{+\infty} \sigma^{-(1+2r/\rho)} F_1(\sigma) d\sigma + \int_1^{+\infty} \sigma^{-(1+\frac{2r}{\rho})} F_2(\sigma) d\sigma < \infty.$$

Consider u_0 as in the proof of Theorem 1.6 (i). Since $\rho < N$, $u_0 \in L^r(\Omega)$. We claim that $\tilde{w}(t) = AS(t)u_0 + 1$ is a supersolution of the problem (1). Indeed, from Lemma 2.4 we conclude that

$\|\tilde{w}(t)\|_{L^\infty} \leq 2AC_0K^{1/r}t^{-\rho/2r}$ for some constant $C_0 > 0$ and $t \in [0, T]$ with T sufficiently small. Arguing as in the derivation of (27) it is possible to conclude

$$\begin{aligned} S(t)u_0 + \int_0^t S(t-\sigma)f(\tilde{w}(\sigma))^{p(\cdot)} d\sigma &\leq S(t)u_0 + \int_0^t S(t-\sigma)\tilde{f}(\tilde{w}(\sigma)) d\sigma \\ &\leq S(t)u_0 + \int_0^t \tilde{G}(\|\tilde{w}(\sigma)\|_{L^\infty})S(t-\sigma)\tilde{w}(\sigma) d\sigma \\ &\leq S(t)u_0 \left(1 + A \int_0^t \tilde{G}(\|\tilde{w}(\sigma)\|_{L^\infty}) d\sigma\right) + \int_0^t \tilde{G}(\|\tilde{w}(\sigma)\|_{L^\infty}) d\sigma \\ &\leq AS(t)u_0 + 1 = \tilde{w}(t) \end{aligned}$$

for $t \in [0, T]$ with T sufficiently small, since

$$\int_0^t \tilde{G}(\|\tilde{w}(\sigma)\|_{L^\infty}) d\sigma = (2AC_0K^{1/r})^{2r/\rho} \left(\frac{2r}{\rho}\right) \int_{t^{-\rho/2r}2AC_0K^{1/r}}^{+\infty} \sigma^{-\left(1+\frac{2r}{\rho}\right)} \tilde{G}(\sigma) d\sigma \rightarrow 0,$$

as $t \rightarrow 0$. Hence, $\tilde{w}$ is a supersolution and we can conclude as in the proof of Theorem 1.6 (i).

4. Uniqueness

Proof of Theorem 1.10.: Suppose that problem (1) has two L^r -solutions u and v defined on some interval $(0, T)$ in the class (8) with initial data u_0 and v_0 respectively. Then,

$$u(t) = S(t)u_0 + \int_0^t S(t-\sigma)f(u(\sigma))^{p(\cdot)} d\sigma \quad (33)$$

and

$$v(t) = S(t)v_0 + \int_0^t S(t-\sigma)f(v(\sigma))^{p(\cdot)} d\sigma. \quad (34)$$

Note that

$$f(u(x, \sigma))^{p(x)} - f(v(x, \sigma))^{p(x)} = p(x)w^{p(x)-1} [f(u(x, \sigma)) - f(v(x, \sigma))],$$

where $w = \theta f(u(x, \sigma)) + (1 - \theta)f(v(x, \sigma))$, $\theta \in [0, 1]$ for a.e. $x \in \Omega$, $0 < \sigma < t < T$. Furthermore, since f is non-decreasing and $u(x, \sigma) \neq v(x, \sigma)$,

$$\left| \frac{f(u(x, \sigma)) - f(v(x, \sigma))}{u(x, \sigma) - v(x, \sigma)} \right| = \frac{f(u(x, \sigma)) - f(v(x, \sigma))}{u(x, \sigma) - v(x, \sigma)} \leq \mathcal{G}(C_1\sigma^{-\rho/2r}).$$

Thus, subtracting (34) from (33), and using estimate (13) we get

$$\begin{aligned} \|u(t) - v(t)\|_{L^r} &\leq \|S(t)(u_0 - v_0)\|_{L^r} + \left\| \int_0^t S(t-\sigma) [f(u(\sigma))^{p(\cdot)} - f(v(\sigma))^{p(\cdot)}] d\sigma \right\|_{L^r} \\ &\leq \|u_0 - v_0\|_{L^r} + p^+ \int_0^t \|S(t-\sigma) \max \{f(u(\sigma))^{p(\cdot)-1}, f(v(\sigma))^{p(\cdot)-1}\} \\ &\quad [f(u(\sigma)) - f(v(\sigma))]\|_{L^r} d\sigma \end{aligned}$$

$$\begin{aligned}
&\leq \|u_0 - v_0\|_{L^r} + p^+ \int_0^t \left\| \frac{f(u(\sigma)) - f(v(\sigma))}{u(\sigma) - v(\sigma)} \right\|_{L^\infty} \\
&\quad \left\| S(t - \sigma) \max \left\{ f(u(\sigma))^{p(\cdot)-1}, f(v(\sigma))^{p(\cdot)-1} \right\} (u(\sigma) - v(\sigma)) \right\|_{L^r} d\sigma \\
&\leq \|u_0 - v_0\|_{L^r} + p^+ \int_0^t \mathcal{G}(C_1 \sigma^{-\rho/2r}) \left\| \max \left\{ f(\|u(\sigma)\|_{L^\infty})^{p(\cdot)-1}, f(\|v(\sigma)\|_{L^\infty})^{p(\cdot)-1} \right\} \right\|_{L^\infty} \\
&\quad \cdot \|u(\sigma) - v(\sigma)\|_{L^r} d\sigma \\
&\leq \|u_0 - v_0\|_{L^r} + p^+ \int_0^t \mathcal{G}(C_1 \sigma^{-\rho/2r}) [f(C_1 \sigma^{-\rho/2r})^{p^+-1} + f(C_1 \sigma^{-\rho/2r})^{p^--1}] \\
&\quad \cdot \|u(\sigma) - v(\sigma)\|_{L^r} d\sigma.
\end{aligned}$$

By condition (9) and Gronwall's inequality we obtain

$$\begin{aligned}
&\|u(t) - v(t)\|_{L^r} \\
&\leq \|u_0 - v_0\|_{L^r} \exp \left[p^+ \int_0^T \mathcal{G}(C_1 \sigma^{-\rho/2r}) [f(C_1 \sigma^{-\rho/2r})^{p^+-1} + f(C_1 \sigma^{-\rho/2r})^{p^--1}] d\sigma \right] \\
&= \|u_0 - v_0\|_{L^r} \exp \left[\frac{2rp^+ C_1^{2r/\rho}}{\rho} \int_{C_1 T^{-\rho/2r}}^\infty \sigma^{-(1+2r/\rho)} \mathcal{G}(\sigma) [f(\sigma)^{p^+-1} + f(\sigma)^{p^--1}] d\sigma \right],
\end{aligned}$$

for $t \in (0, T)$. Hence, the uniqueness follows taking $u_0 = v_0$. ■

5. Conclusion

We consider the parabolic problem $u_t - \Delta u = f(u)^{p(x)}$ in $\Omega \times (0, T)$, with the Dirichlet condition on the boundary of a smooth bounded domain Ω . We assume that $f \in C([0, +\infty))$ is non-decreasing and $p \in C(\Omega, \mathbb{R}_+)$ satisfy $0 < p^- = \min_{x \in \Omega} \{p(x)\} \leq p(x) \leq \max_{x \in \Omega} \{p(x)\} = p^+$. We determine conditions so that this problem admits or not a L^r -solution for non-negative initial data $u_0 \in L^r(\Omega)$ with $r \geq 1$.

In our first result (Theorem 1.3) the existence of a L^r -solution of (1) is guaranteed if either

$$\begin{aligned}
&\limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^+} < \infty \quad \text{and} \quad \limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^-} < \infty, \quad \text{if } r > 1, \quad \text{or} \\
&\int_1^{+\infty} \sigma^{-p^*} F_1(\sigma) d\sigma < \infty \quad \text{and} \quad \int_1^{+\infty} \sigma^{-p^*} F_2(\sigma) d\sigma < \infty, \quad \text{if } r = 1,
\end{aligned}$$

where $F_1(\sigma) = \sup_{1 \leq t \leq \sigma} f(t)^{p^+}/t$ and $F_2(\sigma) = \sup_{1 \leq t \leq \sigma} f(t)^{p^-}/t$. Additionally, we prove that this solutions belongs to the class

$$C([0, T], L^r(\Omega)) \cap \{u \in L_{loc}^\infty((0, T), L^\infty(\Omega)); \sup_{t \in (0, T)} t^{N/2r} \|u(t)\|_{L^\infty} < \infty\}.$$

The non-existence of a L^r -solution is obtained assuming

$$\begin{aligned}
&\limsup_{t \rightarrow +\infty} t^{-p^*} f(t)^{p^-} = +\infty, \quad \text{if } r > 1, \\
&\int_1^{+\infty} \sigma^{-p^*} F_2(\sigma) d\sigma = +\infty, \quad \text{if } r = 1,
\end{aligned}$$

that is, there exist $u_0 \in L^r(\Omega)$, $u_0 \geq 0$ such that problem (1) does not have a L^r -solution.

In our second result, (Theorem 1.6), we obtain new results of existence and non-existence of non-negative L^r -solutions for problem (1). To do this, we consider non-negative initial data $u_0 \geq 0$ in the sets $\mathcal{I}^p(x_0)$ and $\mathcal{I}_\rho(x_0)$. In particular, assuming $f \in C([0, +\infty))$ convex, non-decreasing, $f(0) = 0$,

$$\limsup_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^+} < \infty, \quad \limsup_{t \rightarrow +\infty} t^{-p_0} f(t)^{p^-} < \infty \quad \text{with } p^- \geq 1$$

for some $p_0 > p^* = 1 + 2r/N$ and initial data in $\mathcal{I}^p$ with $0 < \rho < 2r/(p_0 - 1) \leq N$, we obtain a solution in the class

$$C([0, T], L^r(\Omega)) \cap \{u \in L_{loc}^\infty((0, T), L^\infty(\Omega)); \sup_{t \in (0, T]} t^{N/(2\rho)} \|u(t)\|_{L^\infty} < \infty\}.$$

Finally, we establish a conditional uniqueness result in the class where the L^r -solutions obtained are defined, see (Theorem 1.10). This result is more general than the one given in [33].

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